

2024

01

PlasmonicTron

Note: Some sketches and pictures of prototype were omitted due to confidentiality

Copilot

Location

Taiwan

Project Time

05/2024 –
Present

Project Brief

This is a Collaborative project between PlasmonicTron and UC Irvine's biomedical engineering department. It is a microscope system built to monitor retinal organoid growth for regenerative medicine.

My Role

I worked as a product engineer and was involved in the design and engineering of the microscope structure and function, its outer shell, and the tuning of the optical equipment. I worked closely with a mechanical engineer and an optical engineer to ensure all the components fit and are working optimally. In addition, I also helped present past products and Copilot to fair attendees at BioAsia Taiwan 2024.

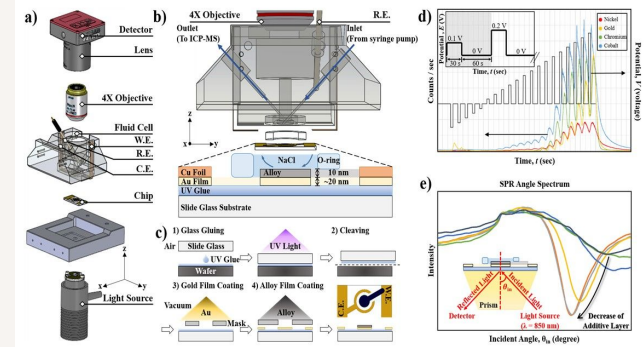


Overview

Imaging Surface Plasmon Resonance (iSPR)

iSPR is a type of label-free biosensor technology that enables real-time and high-throughput detection of biomolecular interactions. The iSPR sensor utilizes the principles of Surface Plasmon Resonance (SPR), which involves the interaction of photons with the surface of a thin metal film (typically gold or silver) and the subsequent detection of changes in the refractive index near the surface.

We have implemented SPR to different system, including the picture in the top right where we implemented it together with electrochemistry and imaging for a chemistry professor at National Taiwan University. In this project, we are using iSPR to examine the metabolic molecules produced by the organoids to determine the state they are in.



Overview

Organoid on-a-chip

Organoid on-a-chip is a micro-fluidic bioreactor system used to standardize retinal organoid (RtOg) production and reduce manual labor in the process. The bioreactor (The small clear chip on the sample tray of the microscope) is designed in AutoCAD, 3D printed, then optimized through in-vitro evaluation. This system provides a controlled environment for RtOg production and can potentially improve the quality and consistency of tissues used in regenerative medicine compared to previous methods.

In Copilot, we are combining this novel method of RtOg production with iSPR to monitor organoids' status by examining the molecular products of the organoids in the micro-fluidic system.

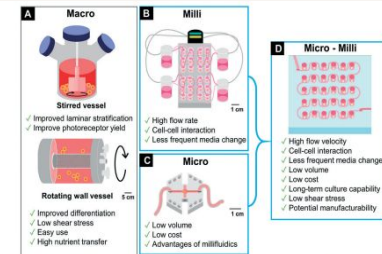


Fig. 1 Review of organoid bioreactors. (A) Macro scale bioreactors: stirred/spinning and rotating wall vessels;¹⁸ (B) Milli/microfluidic bioreactor;¹⁹ (C) microfluidic bioreactor;²⁰ (D) micro-milli/microfluidic bioreactor in this article.

Engineering of Copilot

Mechanical

I worked alongside a mechanical engineer to design and build 2 X-Y translators to motorize the periodic circular motion the organoids need to survive, and the precise movement of the chip under the microscope.

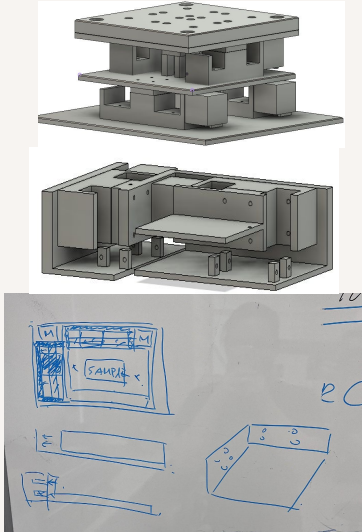
Optical

Due to budget and size constraint for this Copilot prototype, we needed to redesign a parallel light source instead of purchasing directly from Thorlabs. I worked with an optical engineer to test different theories and methods to make parallel light in under 6 meters of space.

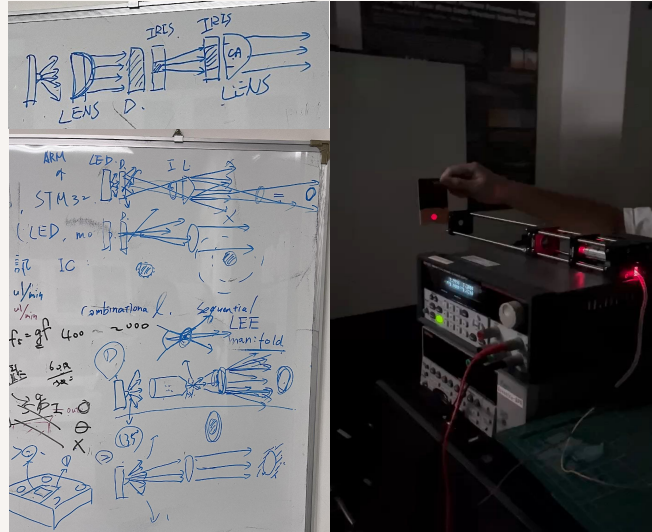
Communication

Working with different engineers for different part of the project proved the importance of order of operation and communication. The margin for error is tightly controlled for systems like this, so each team member must list out their specs before we can define the dimensions for each component of Copilot then proceed.

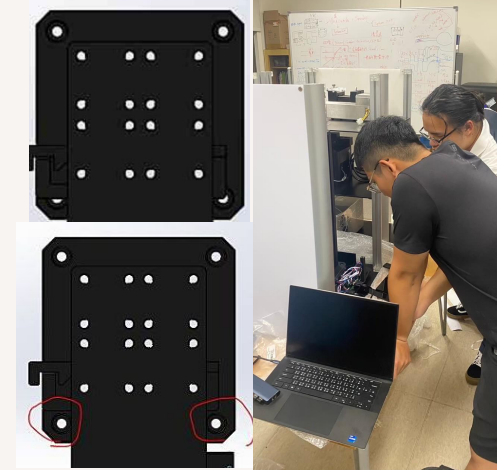
Photos



Top: X-Y translator for the whole system
Middle: X-Y translator for only the sample
Bottom: sketches



Left: Some sketches of methods tested
Right: is the video demonstration of our most successful build for a parallel light source.



Example of one of the errors.
Failure to communicate the build process so there was no space for the screws.

Future and Learning Outcomes

Future

- Continual improvement
 - Fix panel gaps, improve how the parts connect and interact for smoother operation
 - Rework X-Y translator for more powerful motors to ensure smooth and precise movement
 - Communicate with UCI
- Attend MEDICA in Düsseldorf, Germany to see and learn what bigger companies are doing

Takeaways

- Experience prototyping from an engineering perspective
 - Designing isn't just about looks and function, it needs to be physically viable. Ex. weight distribution for the stability of the system
 - The design needs to take into consideration the manufacturing process. Ex. Screw hole placement and the order of construction
- Experience working in an engineering team
 - Working with electrical, mechanical, and optical engineers taught me how important communication and collaboration is in a project
- Communicating with potential customers and stakeholders in big expo and professional settings